

Oleksiy Nedobiychuk

Software Engineer | AI/ML Background | Python Data Pipelines

✉ oleksiy12345@live.it

📧 [@lesi-nedo](https://github.com/lesi-nedo)

📄 [lesi-nedo](https://www.linkedin.com/company/lesi-nedo)

🌐 <http://nedo.im/blog/>

📍 Pisa, Italy

Education

- **M.Sc. in Computer Science (AI curriculum), University of Pisa** [2023 – Present] 109/110(e).
- **B.Sc. in Computer Science, University of Pisa** [2019 – 2023] 106/110.

Relevant Experience

Software Engineer at TD Group Italia S.r.l., Vecchiano, Italy [Jul. 2025 – Present]:

- Built **Python** data pipelines and scheduled processors for sensor/weather data with **APScheduler**, **BeautifulSoup**, and **Requests**, supporting agritech and healthcare IoT platforms.
- Developed data-normalization tools combining **RapidFuzz**, **ONNX** embeddings, **HNSW** indexes, and MySQL mapper tables for semantic matching and source-to-target mapping.
- Improved grapevine models for phenology, maturation, irrigation, evapotranspiration, and disease-risk estimation; created **Highcharts/ECharts** dashboards for model outputs and data-quality signals.
- Maintained full-stack **Django/MySQL** data platforms with **MQTT** ingestion, **Channels/Redis** real-time updates, and **Keycloak**/OIDC role-based access.

Full-Stack Developer at TAI Solutions S.r.l., Pisa, Italy [Sept. – Nov. 2022]:

- Built a **Spring** microservice and **Angular** application for tenant/user management, integrating **Keycloak** authentication and authorization across multiple clients.

Projects

Professional Cycling Analysis – Data Mining

[GitHub](#)

- Cleaned and engineered cycling datasets with imputation, outlier detection, and correlation analysis; applied K-means/DBSCAN clustering and trained a neural network to predict rider performance with 90% test accuracy.

Neural Network for Regression/Classification

[GitHub](#)

- Built a neural network from scratch in **Python/NumPy** with backpropagation, Nesterov momentum, L1/L2 regularization, and grid-search tuning; reached 0.01 MSE on regression and 95% accuracy on classification.

ProbLog Based AI Agent

[GitHub](#)

- Built a probabilistic AI agent for a 2D fighting game, modeling uncertainty with a ProbLog knowledge base and dynamic probability estimation; outperformed MCTS, MinMax, and Prolog baselines in simulation.

Web Application – Full-Stack Development

[Live](#)

- Developed and deployed a **Svelte**/WebGL2 frontend with a **Django**/PostgreSQL backend on Debian using Nginx and Apache, including database design and production server configuration.

Skills

Python/Data Pipelines	• Python, Pandas, NumPy, Scikit-learn, PyTorch, TensorFlow, Matplotlib, Seaborn. • Data pipelines, APScheduler, Requests, BeautifulSoup, CSV/mapper tables, MySQL, PostgreSQL.
ML/Semantic Backend/Tools	• ONNX embeddings, HNSW indexes, RapidFuzz, model evaluation, dashboards. • Django, REST APIs, MQTT, Redis, Docker, Linux, Shell scripting, GIT.
Frontend Languages	• Highcharts, ECharts, Svelte, Angular, TypeScript/JavaScript. • Fluent in English, Italian, and Ukrainian.